KENDRA L. WALKER

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EDUCATION

2010 **Ph.D. in *Natural Resource Policy & Management*, University of Michigan**
School for Environment & Sustainability
Certificate of Graduate Studies, ***Spatial Analysis and GIS***

2000 **B.S., *Ecological, Behavioral & Evolutionary Biology,* University of California at San Diego**

1999 ***Tropical Ecology Field Certificate,* University of Costa Rica**

RESEARCH AND PROFESSIONAL EXPERIENCE

Independent Geospatial Data Consultant 6/2025-present

- Develop custom remote sensing analyses for diverse projects including reduced-emission rangeland management, community-led revegetation, and impacts of climate anomalies on crop yield.
- Work in client-oriented environment, requiring meeting tight deadlines on real-world projects, communicating findings to international partners with ranging technical understanding of geospatial data, and providing tools for long-term implementation.

Remote Sensing Specialist -- Environmental Markets Lab, UC Santa Barbara 2/2021 – 5/2025

- Developed automated forest and land-use change analyses based on optical and radar satellite data to inform complex projects assessing the impact of policies for land-based carbon solutions.
- Built processing pipelines in Python integrating novel and well-established satellite imagery, cutting-edge machine learning models, and gold standard statistical methods.
- Communicated methods and findings to broad community via reports, conferences (including invited NASA talks) and scientific publications, developed training and validation protocols for field data collection in diverse settings, and advanced methods to train and validate geospatial data.

Visiting Scientist -- University of Michigan 2018-19

- Coupled extensive remote-sensing analysis with field data collection to produce the first consistent maps tracking forest-cover change in Panama over three decades.
- Advanced methods of near real-time monitoring of deforestation in tropical forests.

Postdoctoral Researcher -- Smithsonian Tropical Research Institute, Panama 2011-14

- Principal Investigator on study on forest change and greenhouse gas emissions under different forest management strategies. Conducted forest plot inventories across Panama and combined with remote sensing data and historical plot measurements to tune allometric equations and biomass estimates and helped inform baseline estimation for REDD+. Informed research on biomass accumulation in secondary growth and importance of regrowth in meeting national carbon goals.

Spatial Analysis & Modeling Assistant -- IFRI Central African Forestry & Institutions 2009-10
Interdisciplinary team member for NSF-sponsored study of the effects of institutions on forest cover in Africa. Prepared GIS datasets, processed and interpreted high-resolution satellite imagery, informed equation-based and agent-based models, engineered quantitative features from forestry reports in French.

Principal Investigator -- University of Michigan & Wildlife Conservation Society, Gabon 2004-10
Designed and managed research project assessing strategies to protect against crop loss from wildlife in Gabon. Wrote grant proposals, secured funding, trained field assistants, designed and conducted interviews in French, and orchestrated quantitative field assessments and risk mapping in 38 remote villages throughout Gabon.

PUBLICATIONS

Walker, K.L. 2024. Overcoming common pitfalls to improve the accuracy of crop residue burning measurement based on remote sensing data. *Remote Sensing* 16(2) 342. https://doi.org/10.3390/rs16020342

Bardino, G., Di Fonzo, G., **Walker, K.L.**, Vitale, M., Hall, J.S., 2023. Landscape context importance for predicting forest transition success in central Panama. *Landscape Ecology*.

Hall, J.S., Plisinski J, Mladinich, S., van Breugel, M., Lai, H, Asner, G., **Walker, K.L**., Thompson, J. 2022. Deforestation scenarios show the importance of secondary forest for meeting Panama's carbon goals. *Landscape Ecology*.

Walker, K.L. 2021. Effect of land tenure on forest cover and the paradox of private titling in Panama. *Land Use Policy* 109.

Walker, K.L. 2020. Capturing ephemeral forest dynamics with hybrid time-series and composite mapping in the Republic of Panama. *International Journal of Applied Earth Observation and Geoinformation* 87.

Walker, K.L. 2016. Seasonal mixing in forest-cover maps for humid tropics and impact of fluctuations in spectral properties of low vegetation. *Remote Sensing of the Environment* 179 p 80-88.

Walker, K.L. 2011. Labor costs and crop protection from wildlife predation: the case of elephants in Gabon. *Agricultural Economics* 43(1) p. 61-72.

Walker K.L. 2009. Protected-area monitoring dilemmas: A new tool to assess success. *Conservation Biology.* 23(5) p. 1294-1303.

Walker, K.L. 2006. Impact of the Little Fire Ant, *Wasmannia auropunctata*, on native forest ants in Gabon. *Biotropica* 38(5) p.666-673.

Other works

Walker, K.L., Moscona, B., Jack, B.K., Jayachandran, S., Kala N., Pande, R., Xue, J., 2023. Detecting crop burning in India using satellite data. arXiv preprint arXiv:2209.10148.

Jack, B. K. & **Walker, K. L.** 2023. Integrating Remote Sensing and Randomized Control Trials. Collaboration with Conservation International. https://tinyurl.com/RCT-RS-Guide. Modified as chapter to be published in GeoSpatial Impact Evaluation in Practice, AidData (Global Research Institute), 2026.

Walker, K.., Ayala, A., Sharwood, L., & Heilmayr, R., Improving smallholder representation in crop mapping. In review with *Environmental Research: Food Systems.* https://klwalker-sb.github.io/klwalker-sb/files/py_mapping.pdf

Pede, A., **Walker, K**., Ayala, A., Sharwood, L., & Heilmayr, R. Impact of Paraguay's zero deforestation law on forest cover and agricultural systems. In preparation for PNAS

SELECTED PRESENTATIONS

Crop Species Mapping to Understand the Agricultural Impacts of Conservation Policy in Paraguay
Space Week, Nordeste/NASA. Fortaleza, Brazil (in person), August 2023.

Mapping property values to understand land-use change in South America
NASA LCLUC Agricultural Hoptspots Series, May 2023 (co-presented with R. Heilmayr). (link)

Payments for Ecosystem Services to Reduce Crop Residue Burning

GeoField,(AidData), Leveraging Earth Observation for Impact Evaluations of Climate Sensitive Agriculture, Virtual Conference, May 2023. (link)

Panama Vegetation-Cover Time Series Maps – following forest change over three decades in Panama Smithsonian Tropical Research Institute Research Seminar (virtual), March 2021.

LEADERSHIP & TEACHING

Graduate Student Instructor, University of Michigan 2005-06 & 2008-09
Trained over 100 MS/PhD students in fundamental GIS analysis using primarily ArcGIS software. Occasional lecturer for larger 200-student theory-based GIS course. Led discussion groups and study sessions for undergraduate courses (Sustainability and Development, General Ecology).

Presidential Management Fellow Finalist 2010
Status given to outstanding citizen-scholars with exceptional commitment to and excellence in the leadership and management of public policies and programs.

Fulbright Scholar, United States Department of State 2006-07

Graduate Student Coordinator 2004-06
Organized and coordinated meetings for graduate-level interdisciplinary workshop on sustainable livelihoods and conservation. Developed website, facilitated debates and collaborative feedback sessions, invited speakers and managed logistics.

Environmental Educator, Peace Corps, Gabon 2001-04
Collaborated with local community members to develop an environmental education curriculum and materials for six primary schools surrounding Lopé National Park, Gabon. Designed and constructed exhibits for two ecological museums. Authored grant proposals and secured funding for community development projects including a youth center, world environment day, and a pilot ecology camp.

LANGUAGE & COMPUTER SKILLS

English (native), **Spanish** (proficient), **French** (intermediate)

GIS and Remote Sensing analysis in ArcGIS, SAGA QGIS, ERDAS Imagine and Google Earth Engine. Extensive work with PlanetScope, Sentinel, Landsat, MODIS, Aster and Rapid Eye Imagery. Experience with LiDAR, Quickbird, Worldview, SAR (radar), and aerial imagery. Programming in Python, R, Java & Linux Shell (Bash); Parallel Programming for high-performance computing (HPC); Package development and management in GitHub; stable distribution with Docker; Project management and demo in Jupyter; Database design and management in mySQL, MS Access & Excel; Photography and graphics design hobbyist with Adobe Photoshop and Illustrator.